

Decision_Tree

- * DT intuition
- * How to split the nodes?
- * Entropy
- * Building a DT intuition
- * Visualizing the process of building DT
- * Sklearn implementation
- * Gini Impurity
- * Overfit vs. Underfit
- * Hyperparameter Tuning
- * Visualizing DT
- * Feature importance
- * Interview questions
Important terminology
Impact of Outliers
Data Imbalance
Space and time complexity
Using same column twice for split
Using DT for regression

UseCase Intro: Employee Attrition

You are a Data Scientist working at a Jio

- * The company is facing a huge problem of employee attrition
- * Your task is to help the company find a solution to this problem.

Why is attrition a problem?

- * A new employee asks for more compensation
- * Training of new employees
- * Lots of time and resources required for searching a new candidate

<img

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What can be done to solve the problem ?

1. Identify the employees who may leave in future.
 - Targeted approaches can be undertaken to retain such employees.
 - These might include addressing their problems with the company and so on
2. Help identify the key indicators/factors leading to an employee leaving.
 - ##### What all reasons can you think of contributing to attrition ?
Forcing employees to come to office

dailyUnhealthy culture etc

- Identifying these key factors helps in taking better measures to improve employee retention

Dataset

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import io
```

Code

```
!gdown 16KtxSt_QEGQvfluEaMls5cCHPwhRXgCk
```

Output

```
Downloading...
From: https://drive.google.com/uc?id=16KtxSt_QEGQvfluEaMls5cCHPwhRXgCk
To: /content/HR-Employee-Attrition.csv
0% 0.00/228k [00:00<?, ?B/s]100% 228k/228k [00:00<00:00, 5.57MB/s]
```

Code

```
df = pd.read_csv("HR-Employee-Attrition.csv")
df.info()
```

Output

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1470 entries, 0 to 1469
Data columns (total 35 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1470 non-null   int64
1   Attrition                           1470 non-null   object
2   BusinessTravel                       1470 non-null   object
3   DailyRate                           1470 non-null   int64
4   Department                           1470 non-null   object
5   DistanceFromHome                    1470 non-null   int64
6   Education                           1470 non-null   int64
7   EducationField                       1470 non-null   object
8   EmployeeCount                       1470 non-null   int64
9   EmployeeNumber                      1470 non-null   int64
10  EnvironmentSatisfaction              1470 non-null   int64
11  Gender                              1470 non-null   object
12  HourlyRate                          1470 non-null   int64
13  JobInvolvement                      1470 non-null   int64
14  JobLevel                            1470 non-null   int64
15  JobRole                             1470 non-null   object
16  JobSatisfaction                     1470 non-null   int64
17  MaritalStatus                       1470 non-null   object
18  MonthlyIncome                       1470 non-null   int64
19  MonthlyRate                         1470 non-null   int64
20  NumCompaniesWorked                  1470 non-null   int64
21  Over18                              1470 non-null   object
22  OverTime                            1470 non-null   object
23  PercentSalaryHike                   1470 non-null   int64
24  PerformanceRating                   1470 non-null   int64
25  RelationshipSatisfaction             1470 non-null   int64
26  StandardHours                       1470 non-null   int64
27  StockOptionLevel                    1470 non-null   int64
28  TotalWorkingYears                   1470 non-null   int64
29  TrainingTimesLastYear               1470 non-null   int64
30  WorkLifeBalance                     1470 non-null   int64
31  YearsAtCompany                      1470 non-null   int64
32  YearsInCurrentRole                  1470 non-null   int64
33  YearsSinceLastPromotion             1470 non-null   int64
34  YearsWithCurrManager                1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
```

Data Dictionary

Code

```
df.head()
```

Output

```
Age Attrition      BusinessTravel DailyRate      Department \
0  41         Yes      Travel_Rarely    1102              Sales
1  49         No  Travel_Frequently    279  Research & Development
2  37         Yes      Travel_Rarely   1373  Research & Development
3  33         No  Travel_Frequently   1392  Research & Development
4  27         No      Travel_Rarely    591  Research & Development
DistanceFromHome Education EducationField EmployeeCount EmployeeNumber \
0              1              2 Life Sciences              1              1
1              8              1 Life Sciences              1              2
2              2              2      Other              1              4
3              3              4 Life Sciences              1              5
4              2              1      Medical              1              7
... RelationshipSatisfaction StandardHours StockOptionLevel \
0  ...              1              80              0
1  ...              4              80              1
2  ...              2              80              0
3  ...              3              80              0
4  ...              4              80              1
TotalWorkingYears TrainingTimesLastYear WorkLifeBalance YearsAtCompany \
0              8              0              1              6
1             10              3              3             10
2              7              3              3              0
3              8              3              3              8
4              6              3              3              2
YearsInCurrentRole YearsSinceLastPromotion YearsWithCurrManager
0              4              0              5
1              7              1              7
2              0              0              0
3              7              3              0
4              2              2              2
[5 rows x 35 columns]
```

Summary of EDA and Preprocessing

<img

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We perform EDA followed by preprocessing on the data which is covered in the post read

Post Read - Employee Attrition Usecase

Employee EDA: <https://colab.research.google.com/drive/1OdxmAv5q-92ll5Jf8XrDYQp-4-mitgC8?usp=sharing>

Final dataset after preprocessing

Code

```
!gdown 1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
!gdown 10lrWStbl4LizGJh9GJdYmgL70Dgby9n0
!gdown 1nXgVb1M9ST7-N-ry58YfnM4jrbLfnctk
!gdown 1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
```

Output

```
Downloading...
From: https://drive.google.com/uc?id=1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
To: /content/preprocessed_X_sm.joblib
100% 414k/414k [00:00<00:00, 7.20MB/s]
Downloading...
From: https://drive.google.com/uc?id=10lrWStbl4LizGJh9GJdYmgL70Dgby9n0
To: /content/y_sm.joblib
100% 30.3k/30.3k [00:00<00:00, 59.0MB/s]
Downloading...
From: https://drive.google.com/uc?id=1nXgVb1M9ST7-N-ry58YfnM4jrbLfnctk
To: /content/X_test.joblib
100% 86.6k/86.6k [00:00<00:00, 4.62MB/s]
Downloading...
From: https://drive.google.com/uc?id=1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
To: /content/y_test.joblib
100% 12.6k/12.6k [00:00<00:00, 30.2MB/s]
```

Code

```
import joblib
X_train = joblib.load('preprocessed_X_sm.joblib')
X_test = joblib.load('X_test.joblib')
y_train = joblib.load('y_sm.joblib')
y_test = joblib.load('y_test.joblib')
```

Code

```
# train data shape
X_train.shape
```

Output

```
(1848, 33)
```

Code

```
# test data shape
X_test.shape
```

Output

```
(368, 33)
```

Code

```
X_train.head()
```

Output

```
Age  DailyRate  DistanceFromHome  Education  EducationField  \
0    36         1174                3         4         0.233869
1    21          546                5         1         0.127479
2    43          422                1         3         0.151584
3    42          188               29         3         0.127479
4    35          992                1         3         0.127479
EnvironmentSatisfaction  Gender  HourlyRate  JobInvolvement  JobLevel  ...  \
0                        1         0         99              3         2  ...
1                        3         1         97              3         1  ...
2                        4         0         33              3         2  ...
3                        2         1         56              1         2  ...
4                        4         1         68              2         1  ...
YearsAtCompany  YearsInCurrentRole  YearsSinceLastPromotion  \
0                        5         4                        0
1                        2         2                        2
2                        5         2                        2
3                        1         0                        0
4                        3         0                        1
YearsWithCurrManager  BusinessTravel_Travel_Frequently  \
0                        1                        False
1                        2                        False
2                        2                        True
3                        0                        False
4                        2                        False
BusinessTravel_Travel_Rarely  Department_Research & Development  \
0                        True                        False
1                        True                        True
2                        False                       True
3                        True                        True
4                        True                        True
Department_Sales  MaritalStatus_Married  MaritalStatus_Single
0                        True            False            True
1                        False            False            True
2                        False            True             False
3                        False            False            True
4                        False            False            True
[5 rows x 33 columns]
```

DT intuition

Say, we have attrition data with 2 features

- Age
- Overtime

When we plot the data, it looks something like this :

<img

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Can we use logistic regression to classify this data ?

No as it is a linear model and we have non linear data with us.

* Logistic regression will not perform well in this case.

Can we use KNN to solve this problem ?

Yes as it works on assumptions that

- neighbourhood should be homogenous
- i.e. datapoints belonging to same class are close to each other.

This assumption is valid here.

But, KNN has a big disadvantage

* We cant productionize KNN for large datasets

Why?

Because entire training data is used every time we make an inference.

So, it will be difficult to use KNN in this case.

Lets try something else.

Lets try to split the data into regions using axis parallel hyperplanes

The splitted region along with decision boundaries will look like :

<img

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With the help of these hyperplanes,

- we can represent these homogenous splitted region using if else conditions

Lets see how it looks like :

<img

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Now, that we have the conditions with us

Lets try to represent these conditions digramatically:

Notice that

- This is resembling a tree structure

where

- top most node is called root node
- Bottom most node is called leaf node
- and the nodes in between them are called Internal node

This is called a Decision Tree

Visualization

Heres an visualization to help you understand how we use these axis parallel hyperplane to split the data.

<https://mlu-explain.github.io/decision-tree/>

Notice that

- how easily it is able to classify multiclass data

* which might have been an overhead if we decided to use logistic regression

One of the main advantage of Decision Trees are

DTs are easily interpretable

How so?

* Lets take an example:

One conclusion that can be derived from above decision tree is

Employees younger than 29 yrs and work overtime ≥ 2.5 hrs are more likely to churn This can help the company take specific decisions in this direction to reduce attrition such as:

Incentivise overtime Reducing workload

<img

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If you notice that,

- while building the hyperplanes here
- we were trying to split data into homogenous region/ subspace containing only single class datapoints

What do we need to create these homogenous subspace ?

We need 2 things

- Feature
- Condition to split the data.

We have the features with us.

- We just need to figure out how to find these split conditions.

This is what learning/ training means in decision tree

Lets learn how to split the data into homogenous regions

Points to Remember

<img

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How to split the nodes ?

But, how do we find these rules ?

- * Obviously, we cannot manually create rules for the entire data
- * So, we will have to learn these rules from the data

What is the end goal of classification ?

Our purpose is to achieve max possible prediction confidence and accuracy

<img

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How do decision trees improve prediction confidence ?

* By dividing heterogenous subspaces into smaller subspaces

We want

- A homogenous subspace i.e. which contains data belonging to only one class

Why ?

- Confidence of prediction will be maximum in this case

Say, we have a node with following distribution

And we are given with two options to split this node.

<img

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Which option will give us more homogenous (purer) nodes ?

...

Given the following split options for splitting root node,

Root node : 100 +ve, 100 -ve datapoint

Option 1:

Left node : 10 +ve, 90 -ve datapoints

Right node : 90 + ve, 10 -ve datapoints

Option 2 :

Left node : 60 +ve, 40 -ve datapoints

Right node : 40 +ve, 60 -ve datapoints

Which option will give us more homogenous nodes ?

Correct option: Option 1

...

But, why are we looking for pure nodes ?

In order to understand that,

Lets see how DT assign class label at prediction time.

During prediction time,

- it predicts the label as the one of majority class at that node.

Lets see with an example:

As we are taking majority vote

- pure node will have more confidence in the prediction

If we were to pick option 1

- We have purer node and more homogenous data
- Itll give us more confidence in our prediction

If we were to pick option 2

- There is less condifence in the prediction
- i.e. we are 60% confident in our prediction (left child node) that our datapoint belong to positive class
- Hence, node is impure and more hetrogenous data

So, well pick option 1 for split.

But, what does confidence here means ?

Think of condifence as class probability.

For example:

- In left node,
- we have datapoints as follows:

```
+ve class : 10  
-ve class : 90
```

So, we can say that

- if a datapoint belong to left node
- there is a 90% probability it belongs to -ve class

or class probability is 0.9.

Conclusion:

- Purer the nodes, more homogenous the data is.
- More homogenous the data in the node is, more confident we are about our prediction belonging to specific class.
- purer the node, more the probability that datapoints belongs to particular class.

...

Quiz- 6 - Check your understanding

Which of the following statement is false ?

- a. Purer the node, more confidence we are in our prediction
- b. For making prediction, DT takes majority vote of class.
- c. More homogenous the data is at the node, more confident we are about our prediction
- d. None of the above

Correction option: d. None of the above

...

So, our objective is:

- To split a node in such a way
- that we get purer nodes

But, how do we quantify the purity of the nodes ?

How do we calculate purity mathematically ?

We do so using Entropy

Lets understand what entropy is

Entropy

Entropy is used to measure the impurity not purity

- i.e. it measures the heterogeneity of the node.

Fun fact:

- The concept of entropy comes from basic probability/ information theory
- where it is used to measure the randomness.

So,

More the heterogeneity in the node,

- larger the value of entropy will be and vice versa.

...

Quiz 7 - Try it yourself

In which of the cases will the entropy be minimum ?

- A. A node will all datapoints belonging to one class only
- B. Node with datapoints belonging to both class
- C. Node with 100 datapoints with one datapoint belonging to positive and rest belonging to other class
- D. Entropy is independent of proportion of datapoints in the node.

Ans: A. A node will all datapoints belonging to one class only

...

Since we want the nodes to be pure,
- we want entropy as low as possible.

Lets look into the fomulation of Entropy

<img

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Entropy formulation

Say, Y be a discrete random variable.

* it can take k discrete values i.e $y \in \{y_1, y_2, y_3, \dots, y_k\}$

How many discrete values of y do we have in employee attrition problem ?

Since, there are two classes (churn or stay),
- we have two discrete values for y in this case.

What will be the entropy for our binary case classification problem ?

<img

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Lets try to understand entropy for binary class case using an example

Say, we have 4 jars.

- Each jar is filled with balls

Balls are of two colors

- Red ball
- Blue ball

You have been asked to pick a ball from these 4 jars

Which jars will give you highest confidence on whether the ball you have picked is red or not ?

Jar 3 and 4 will give us highest confidence.

- As there jars are pure.
- So, we are pretty sure
- We wont get a red ball in Jar 3
- And well definately get red ball in Jar 4.

* In Jar 1,

which is impure (as it has both red and blue balls)we wont have much confidence on whether the ball being picked is red or not.

Lets represent these confidence in numerical terms. i.e. calculate entropy of each jar.

<img

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Notice that

- Jar with equal number of red and blue balls have highest entropy i.e. 1
- Jar with only red balls or blue balls has lowest entropy i.e. 0.

...

Quiz-8

Which of the following statements is true ?

- More the entropy, more pure the node is.
- Less the entropy, more impure the node is.
- More the entropy, more homogenous the node is.
- Less the entropy, more homogenous the node is.

Correct option: d. Less the entropy, more homogenous the node is

Explanation:

- More pure the node is i.e. more homogenous

- less will be the entropy.

...

...

At what probability value will the entropy value be maximum for binary class?

- A. 0
- B. 0.5
- C. 0.33
- D. 1

Ans: B. 0.5

...

Lets understand it by plotting the the curve

Plotting entropy

Lets plot entropy for binary system and see how it looks like:

Desmos plot: <https://www.desmos.com/calculator/avaplvktso>

Code

```
from IPython.display import IFrame
```

Code

```
IFrame(src="https://www.desmos.com/calculator/avaplvktso", width=700, height=375)
```

Output

```
<IPython.lib.display.IFrame at 0x7f30bc437af0>
```


Notice that

- Value of entropy is maximum
- when the probability is 0.5 i.e. equal number of datapoints for each class
- Entropy value is minimum (Entropy = 0)
- when the probability $P(Y = 1)$ is either 0 or 1 (pure node)

Visualization for entropy

In case you want to see how entropy changes as number of datapoint changes, heres a visualization for that.

<https://mlu-explain.github.io/decision-tree/>

Conclusion:

- Entropy is maximum when the node is impure ($P(Y = 1) = 0.5$) i.e. 1
- Entropy is minimum when the node is pure i.e. 0
- Entropy lies between 0 and 1.

Now, that we have learnt about purity and how to use entropy to calculate that.

Lets see how Decision Trees uses it for internal working

Points to Remember

<img

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How Decision Tree works? Building a DT intuition

Lets understand it using a dummy example.

Consider a dataset with 100 datapoints

- and 2 features (Gender , Age < 35)

<img

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Can we use this root node for prediction ?

No. We cant. It is highly hetrogenous

Hence, itll have a high entropy.

Lets calculate its entropy

<img

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So, we should split the node in order to reduce the entropy.

- and make it homogenous.

We have two features with us. i.e. Gender and Age < 35.

Which feature shall we use for splitting ?

We dont know until we calculate the entropy of split for the feature.

Splitting using Gender feature and age<35

Lets first split using Gender feature

When we split using gender

- we get two child nodes
- one for Male
- other for Female.

<img

src=https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/040/798/original/Screenshot_2023-07-28_at_11.14.34_AM.png?1690525348 width =700>

Lets calculate the entropy of child to check whether the entropy has reduced or not

Now that we have calculated entropy of parent as well as both child

- we need to check if there is a reduction in entropy.

But, each child node has an entropy value.

- In order to see the reduction in entropy,
- we first need to combine the child entropy to get a single value out of it.
- then we can compare it with parent entropy.

How do we combine child entropy ?

...

Quiz 11 - Try it out

Which strategy should we use for combining child entropy?

- A. Simple average
- B. Weighted average
- C. Median of child entropy
- D. Maximum of child entropy

Ans: B. Weighted average

...

Simple average?

* When we take simple average,
we are ignoring the proportions of datapoints belonging to each node

There can be a case where

- Child 1 contains 98 datapoints
- Child 2 contains 2 datapoints

We should include the number of datapoints it is impacting while calculating combined entropy.

So, we should take weighted average in this case.

How do we calculate weighted entropy of child nodes ?

We simply do so by multiplying the datapoint proportion with its entropy value

<img

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There is a slight reduction in weighted child entropy (0.88) compared to parent entropy (0.97)

* So, we are moving towards purer nodes.

This reduction in entropy i.e. Parent - weight entropy of child is termed as Information gain

<img

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We can say that we want to

- maximize information gain
- or minimize entropy

We want to maximize information gain.

So, there is chance that there is some other feature

- which is providing more information gain than Gender feature

So, we should use that feature instead.

This means we should calculate Information gain for Age < 35 feature as well

...

Quiz 12 - Check your understanding

Which of the following statement related to Information gain is false ?

- a. Information gain is defined as reduction in entropy.
- b. We want to maximize information gain
- c. It is calculated by subtracting weighted child entropy from parent entropy.
- d. It is calculated by subtracting parent entropy from weighted child entropy

Correct option : d. It is calculated by subtracting parent entropy from weighted child entropy

...

The information gain for each feature is as follows:

$$* I_G(\text{Parent}, \text{Age} < 35) = 0.257$$

$$* I_G(\text{Parent}, \text{Gender}) = 0.091$$

Which feature shall we pick to split the root node ?

We pick feature s.t it gives us maximum information gain

- In this case, splitting using Age < 35 is giving us maximum information gain.
- So, we'll pick it to split our root node.

Lets split our node using Age<35 feature

<img

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Lets talk about left child node

Has the entropy of left child node reduced to 0 or close to 0 ?

No.

- Although the entropy is lower than before
- but it is still an impure node.

In order to achieve more confidence in prediction,

- we should further split this node.

Again, in order to split this node

- well calculate information gain using features
- say, these features are gender, salary, years of experience.

Whichever feature gives us the maximum information gain,

- well split the node using that feature.

Assume that gender gave us highest info. gain,

- well split this node using gender

<img

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We continue doing so

- until we get purer nodes
- i.e. confidence in prediction is high.

(In case learners ask) What happens if we have more than 2 categories for a feature ? How do we split in that case ?

In case where there are more than 2 categories in a feature,

- we simply make a child node for each category

<img

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Now we have seen how Decision Tree works

Lets implement it using sklearn

Sklearn implemenation

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Fetching data

Code

```
!gdown 1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
!gdown 10lrWStbl4LizGJh9GJdYmGL70Dgby9n0
!gdown 1nXgVb1M9ST7-N-ry58YfnM4jrBLfnctk
!gdown 1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
```

Output

```
Downloading...
From: https://drive.google.com/uc?id=1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
To: /content/preprocessed_X_sm.joblib
100% 414k/414k [00:00<00:00, 7.01MB/s]
Downloading...
From: https://drive.google.com/uc?id=10lrWStbl4LizGJh9GJdYMgL70Dgby9n0
To: /content/y_sm.joblib
100% 30.3k/30.3k [00:00<00:00, 48.4MB/s]
Downloading...
From: https://drive.google.com/uc?id=1nXgVb1M9ST7-N-ry58YfnM4jrblFnctk
To: /content/X_test.joblib
100% 86.6k/86.6k [00:00<00:00, 4.56MB/s]
Downloading...
From: https://drive.google.com/uc?id=1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
To: /content/y_test.joblib
100% 12.6k/12.6k [00:00<00:00, 35.2MB/s]
```

Code

```
import joblib
X_train = joblib.load('preprocessed_X_sm.joblib')
X_test = joblib.load('X_test.joblib')
y_train = joblib.load('y_sm.joblib')
y_test = joblib.load('y_test.joblib')
```

Reading data

Importing sklearn DT

sklearn DT doc - <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>

Code

```
from sklearn.tree import DecisionTreeClassifier
```

Initiating instance

Code

```
model = DecisionTreeClassifier(criterion='entropy')
```

Fitting the model

Code

```
model.fit(X_train, y_train)
```

Output

```
DecisionTreeClassifier(criterion='entropy')
```

Get the train / test score

Code

```
model.score(X_test, y_test)
```

Output


```
0.7527173913043478
```

Code

```
model.score(X_train, y_train)
```

Output

```
1.0
```

...

Quiz 14 - Check your understanding

What does the following model performance say about the model ?

Train accuracy - 1.0

Test accuracy - 0.75

Options

- a. Model is underfitting
- b. Model is overfitting
- c. Model is best fit model
- d. None of the above

Correct option: b. Model is overfitting

...

There is a huge difference in train and test score

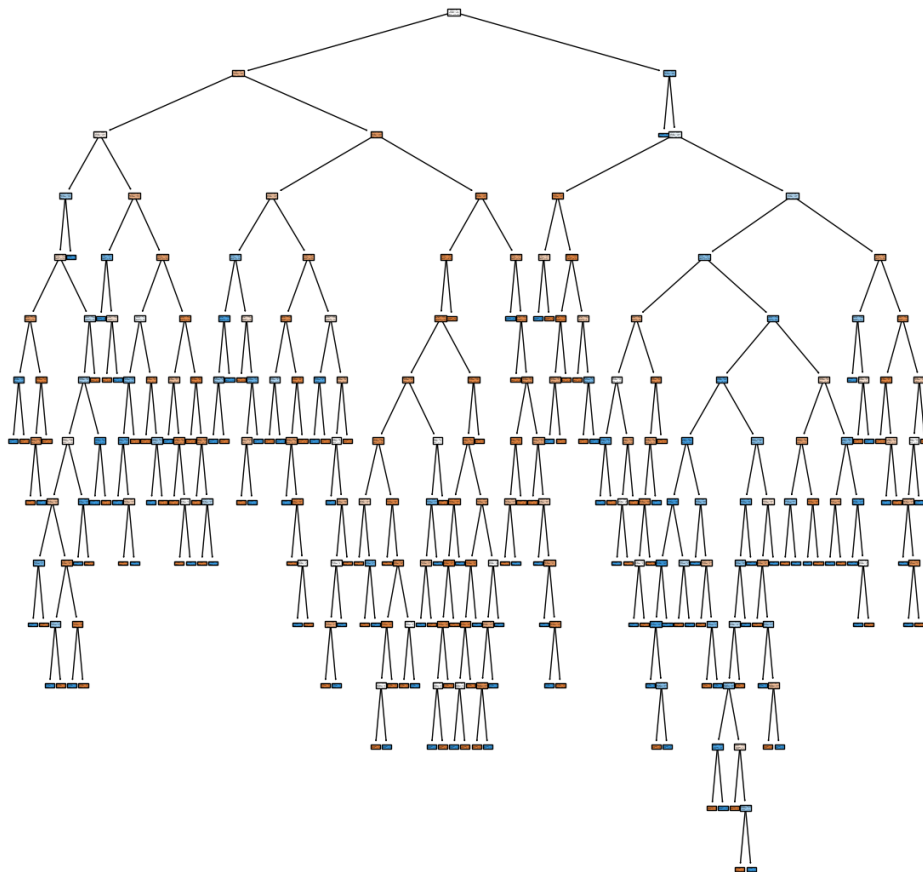
- Model is currently overfitting.

Visualizing DT

Code

```
from sklearn.tree import plot_tree
plt.figure(figsize=(14,14))
plot_tree(model, filled = True);
```

Output Image



Notice that
- how deep is this DT is

How do we stop it from overfitting ?

We'll learn about how to handle bias variance tradeoff in DT in next lecture

Points to Remember

<img

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<img

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Visualizing the process of building DT

link: https://drive.google.com/file/d/1CBFhQ2bYf81kTN-x_T6B1hcbIVqbXjAz/view?usp=sharing

paper: [https://opus.bibliothek.uni-](https://opus.bibliothek.uni-augsburg.de/opus4/frontdoor/deliver/index/docId/79711/file/ECML_PKDD_Decision_Tree_Learning.pdf)

[augsburg.de/opus4/frontdoor/deliver/index/docId/79711/file/ECML_PKDD_Decision_Tree_Learning.pdf](https://opus.bibliothek.uni-augsburg.de/opus4/frontdoor/deliver/index/docId/79711/file/ECML_PKDD_Decision_Tree_Learning.pdf)

Steps to follow:

1. Unzip the file
2. Launch index.html to launch the project
3. Go to Data (top right) -> Import training data -> data.csv

1. Dataset are present in folder named Files. You can move your custom dataset into that folder. However, there are few limits on number of columns and format of data (csv)

It contains data for attrition use case (3 features)

- Gender
- Age < 35
- Marital Status

1. Select Mode as Stepwise. It'll build DT node by node.
2. Click on build.
3. Click build again to move to next stage of DT

For each stage, you can view the information gain on left pane of window.

Gini Impurity

Similar to entropy, Gini is also

- A measure of impurity of nodes.

Optional: Why Gini Impurity ?

Issue with Entropy

Do you recall how entropy is calculated?

- $H(y) = - \sum_{i=1}^k p(y_i) \log(p(y_i))$

<img

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\

So, whats the problem?

- Notice that we take log of probability.
- We have to do this for each feature, at each node.

This is computationally expensive and time consuming.

\

Thats when Gini Impurity comes into picture.

But how do we calculate Gini Impurity?

<img

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Visualizing Gini Impurity

Desmos plot: <https://www.desmos.com/calculator/yhcwubphxs>

Code

```
from IPython.display import IFrame
IFrame(src="https://www.desmos.com/calculator/yhcwubphxs", width=700, height=375)
```

Output

```
<IPython.lib.display.IFrame at 0x7f1482a5a110>
```

How do we calculate Information Gain for Gini Impurity?

$IG = GI_{\text{parent-node}} - \text{Weighted-}GI_{\text{child-nodes}}$

<img

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How to split on numerical features?

I hope you remember that our data contains both,

- Categorical features
- Numerical features

\

We already know that for Categorical feature,

- DT splits data based on distinct categories.

<img src=

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What happens when we have a Numerical feature?

DT uses a threshold-based approach.

* Consider a numerical feature f_1 with n different values.

* We select a threshold value (say n_i) that splits the data into two subsets.

* This is done on the basis of a condition such as:

$f_1 < n_i$ $f_1 > n_i$ $f_1 = n_i$

\

But how do we choose the threshold?

The goal is to find the $\text{threshold+condition}$ that gives us maximum information gain after splitting.

<img

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1. Arrange the values in x_1 in increasing order
2. Set each value of x_1 i.e. (x_1 , x_2 , x_3 ,...) as threshold
3. Calculate the IG of the split for all values

<img

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1. Which gives us n IG values say IG_1 , IG_2 , IG_3 IG_n
2. Choose whichever threshold gives the highest IG.

Do you see any problem in splitting on numerical features?

\

Time Complexity

- For a feature with 1000 different values,
- We have to compute 1000 different splits.
- This is very inefficient.

\

How can we make it more efficient?

By binning the feature values.

- Suppose we have an column.
- Range: [20, 60]

<img

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* We can bin the values into .

Bins: [20, 30), [30, 40), [40, 50), [50, 60]

Overfit Vs Underfit

Recall when we calculated the train & test accuracy for our DT -

* Accuracy -

Train: 1.0 Test: 0.76

<img

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\

What can we conclude from this?

- * There's a big difference in training & test accuracy.
- * It means, the DT model is overfitting the data.

\

When do you think a DT will overfit?

If all the leaf nodes are pure then,

- the DT would have 100% accuracy on training set.

\

Why does DT overfit as the depth increases?

Imagine we have 1000 data points.

- as the depth increases, the data set becomes smaller.

By the time we reach the leaf node,

- there will be fewer points.
- these points could be noise/outliers.

<img

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This results in Overfitting.

...

Quiz 2:

Which of the following statements is true about overfitting in decision trees?

- A. It leads to high bias and high variance.
- B. It leads to low bias and high variance.
- C. It leads to high bias and low variance.
- D. It leads to low bias and low variance.

ANS: B. It leads to low bias and high variance.

...

How can we resolve this problem?

- * Stop splitting nodes before they become pure.
- * Or in simpler terms, by reducing the Depth of the DT.

...

QUIZ 3:

What is pruning in decision trees?

- A. The process of removing outliers
- B. The process of handling missing values
- C. The process removing unnecessary branches
- D. process of transforming categorical variables

ANS: C. The process removing unnecessary branches.

...

But what if the depth is too low?

\

Typically, we let the tree grow until we get a pure node.

What if we stop the tree growth at a depth=2?

This would mean

- Lot of -ve points might also fall in that leaf node,
- and theyll be incorrectly classified as +ve.

\

This results in Underfitting.

\

If the tree depth is less,

- it means the splits are less.

Hence, number of hyper-planes dividing the data space are less </br> i.e., less variance and more bias.

Extreme Case

...

QUIZ 4:

What if the tree depth is zero?

Itll mean that the tree has

- A. a single node
- B. at most 2 nodes

C. at most 3 nodes

ANS: A. a single node

...

Conclusion:

- * A deep tree usually Overfits the training data.
- * A shallow tree usually Underfits the training data.

<img

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Tradeoff

How do we find the right depth of our DT?

Our goal is to -

- not overfit outliers
- not underfit training samples

\

The idea here is to -

- treat depth as a hyperparameter
- and find its optimal value.

Hyperparameters

Lets look at various DT hyperparameters available in the sklearn library.

sklearn DT - <https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html>

What does each of these hyperparameters do?

\

- Criterion:
- Defines which impurity measure to use.
- By default, it is set to Gini Impurity.

* max_depth:

Defines maximum depth upto which a tree will grow. By default, it is None i.e. itll grow until all leaves are pure.

* min_samples_split:

Defines the minimum no. of datapoints required to split further.Helps control depth which therefore prevents overfit.By default, it is 2.

* min_samples_leaf:

Defines the minimum no. of samples a leaf node can have.

* max_leaf_nodes:

Defines the maximum no. of leaf nodes a tree can have.

* max_features:

Selects features to be used while deciding the best split.By default, it considers all the features for split.

* class_weight:

Assign weights to different classes during training.Helps in dealing with imbalanced data.

Tuning the **max_depth**** hyperparameter to improve the model performance.**

\

We try multiple values of depth.

- Train DTs using various depth values.
- Check the CV error for each value.
- Choose the one which gives lowest CV error.

Code

```
import pandas as pd
import numpy as np
```

Code

```
!gdown 1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
!gdown 10lrWStbl4LizGJh9GJdYmGL70Dgby9n0
!gdown 1nXgVb1M9ST7-N-ry58YfnM4jrbLfnctk
!gdown 1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
```

Output

```
Downloading...
From: https://drive.google.com/uc?id=1ZVAP0iLFgrcRZUssN4qPLMQ7NgIj9i8q
To: /content/preprocessed_X_sm.joblib
100% 414k/414k [00:00<00:00, 7.74MB/s]
Downloading...
From: https://drive.google.com/uc?id=10lrWStbl4LizGJh9GJdYMgL70Dgby9n0
To: /content/y_sm.joblib
100% 30.3k/30.3k [00:00<00:00, 60.4MB/s]
Downloading...
From: https://drive.google.com/uc?id=1nXgVb1M9ST7-N-ry58YfnM4jrbLfncTk
To: /content/X_test.joblib
100% 86.6k/86.6k [00:00<00:00, 4.28MB/s]
Downloading...
From: https://drive.google.com/uc?id=1YIAimyBw2AjdAXAt0UU52jLX0vPjS5TG
To: /content/y_test.joblib
100% 12.6k/12.6k [00:00<00:00, 22.3MB/s]
```

Code

```
import joblib
X_train = joblib.load('preprocessed_X_sm.joblib')
X_test = joblib.load('X_test.joblib')
y_train = joblib.load('y_sm.joblib')
y_test = joblib.load('y_test.joblib')
```

Code

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import KFold, cross_validate, cross_val_score
kfold = KFold(n_splits=10)
depths = [3,4,5,6,7,9,11,13,15]
for depth in depths:
    tree_clf = DecisionTreeClassifier(random_state=7, max_depth=depth)
    cv_acc_results = cross_validate(tree_clf, X_train, y_train, cv = kfold, scoring =
    'accuracy', return_train_score = True)
    print(f"K-Fold for depth:{depth} Accuracy Mean: Train:
    {cv_acc_results['train_score'].mean()*100} Validation:
    {cv_acc_results['test_score'].mean()*100}")
    print(f"K-Fold for depth: {depth} Accuracy Std: Train:
    {cv_acc_results['train_score'].std()*100} Validation:
    {cv_acc_results['test_score'].std()*100}")
    print('*****')

```

Output

```

K-Fold for depth:3 Accuracy Mean: Train: 78.31892302488552 Validation: 75.10663924794359
K-Fold for depth: 3 Accuracy Std: Train: 0.9307719004403879 Validation: 4.554589537795214
*****
K-Fold for depth:4 Accuracy Mean: Train: 82.10081048499006 Validation: 76.19242068155113
K-Fold for depth: 4 Accuracy Std: Train: 1.1579051752971248 Validation: 4.501237256858836
*****
K-Fold for depth:5 Accuracy Mean: Train: 85.90672917919422 Validation: 79.60458284371327
K-Fold for depth: 5 Accuracy Std: Train: 0.6957962075134733 Validation: 4.080000122447316
*****
K-Fold for depth:67 Accuracy Mean: Train: 100.0 Validation: 82.8575205640423
K-Fold for depth: 67 Accuracy Std: Train: 0.0 Validation: 7.786460184927287
*****
K-Fold for depth:9 Accuracy Mean: Train: 97.11404753920164 Validation: 82.80141010575794
K-Fold for depth: 9 Accuracy Std: Train: 0.3521803004376563 Validation: 7.105874162015295
*****
K-Fold for depth:11 Accuracy Mean: Train: 98.95384268467551 Validation: 82.80258519388953
K-Fold for depth: 11 Accuracy Std: Train: 0.4008012831162821 Validation: 7.490493735360857
*****
K-Fold for depth:13 Accuracy Mean: Train: 99.75348290277071 Validation: 82.53260869565217
K-Fold for depth: 13 Accuracy Std: Train: 0.23818007715094902 Validation: 8.252281195862817
*****
K-Fold for depth:15 Accuracy Mean: Train: 99.95189416716777 Validation: 82.8575205640423
K-Fold for depth: 15 Accuracy Std: Train: 0.07977449886603395 Validation: 7.835090351040905
*****

```

Training with the best parameters

Code

```
tree_clf = DecisionTreeClassifier(random_state=7, max_depth=4)
tree_clf=tree_clf.fit(X_train, y_train)
pred = tree_clf.predict(X_test)
```

How to ****visualize**** a DT?

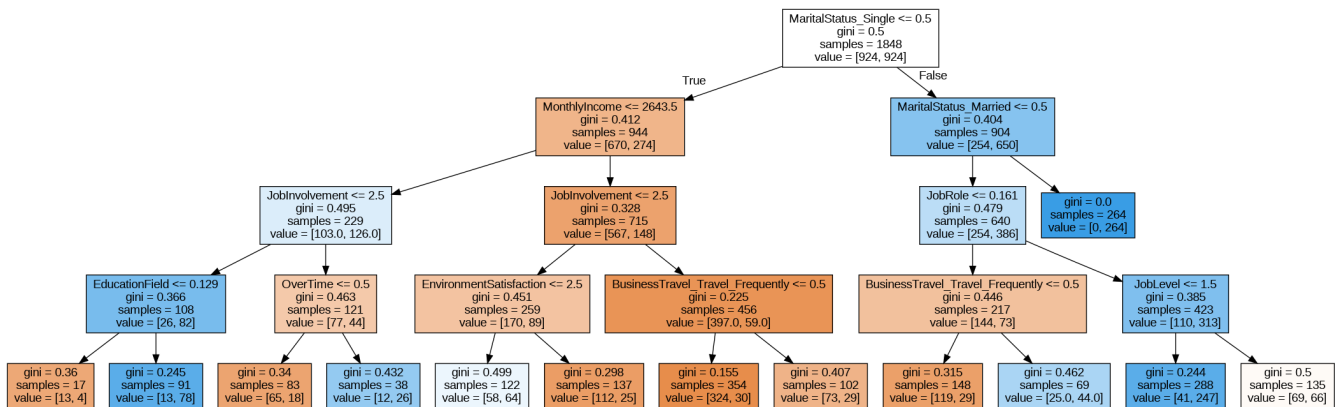
Code

```
from IPython.display import Image
from six import StringIO
from sklearn.tree import export_graphviz
import pydot
features = list(X_train.columns)
# Ensure the target variable 'Attrition' is not there in the feature list
```

Code

```
dot_data = StringIO()
export_graphviz(tree_clf, out_file=dot_data, feature_names=features, filled=True)
graph = pydot.graph_from_dot_data(dot_data.getvalue())
Image(graph[0].create_png())
```

Output Image



Did you notice the depth of the tree?

- Neither too large
- Nor too low

\

What can we infer from this?

If you notice the leaves,

- not all of them are pure.
- This means lesser variance.

Confusion Matrix

Plotting the confusion matrix for test data -

Recall the TP, TN, FN, FP we discussed earlier.

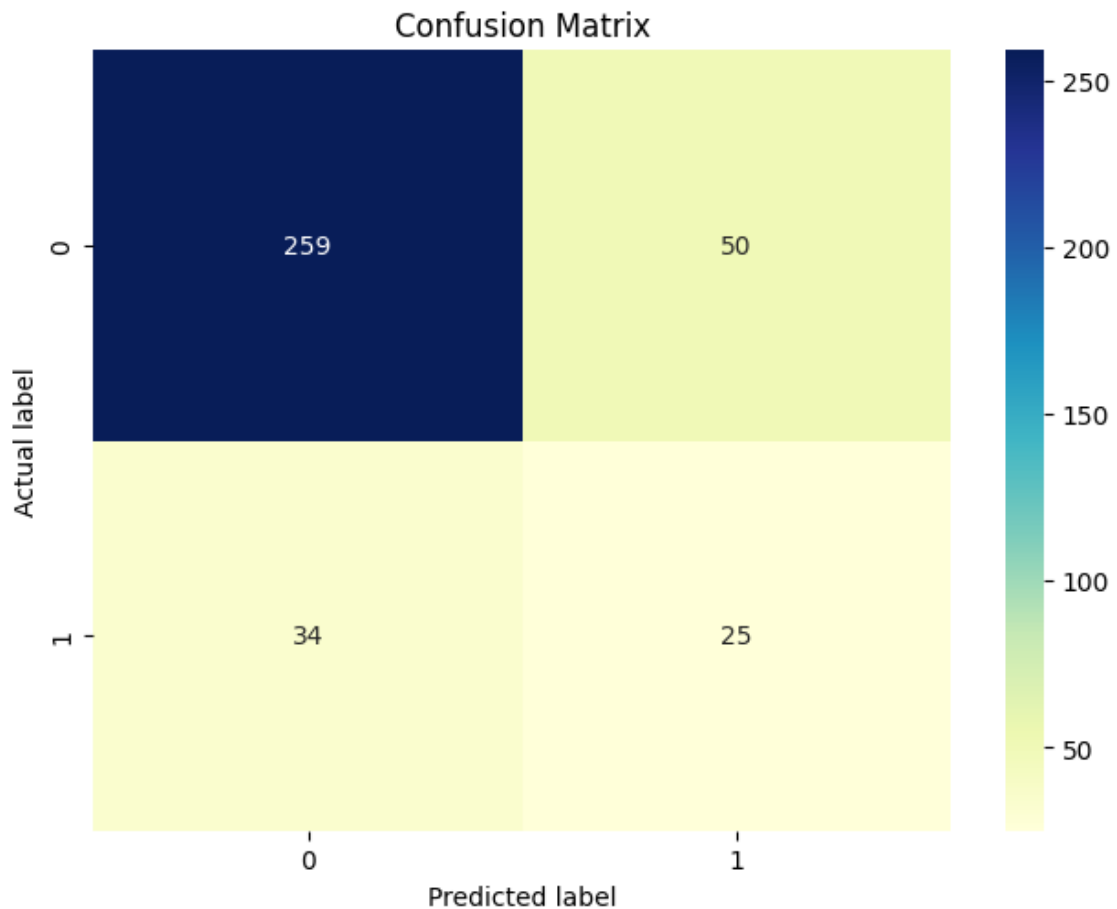
Code

```
# Confusion Matrix
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
cnf_matrix = confusion_matrix(y_test, pred)
fig, ax = plt.subplots()
# create heatmap
sns.heatmap(pd.DataFrame(cnf_matrix), annot=True, cmap="YlGnBu" ,fmt='g')
plt.tight_layout()
plt.title('Confusion Matrix')
plt.ylabel('Actual label')
plt.xlabel('Predicted label')
```

Output

```
Text(0.5, 23.52222222222222, 'Predicted label')
```


Output Image



What did you notice?

- * Model is classifying class 0 quite well.
- * However, the number of FN is large.

Feature Importance

How can we find which parameters are contributing more towards a prediction?

- * This is where feature importance comes into picture.

\

How can we get Feature Importance for DT?

Suppose we have 10,000 datapoints at root node.

- A feature f_i is used twice in splitting a DT.
- At first split,
- No. of data points be 5,000 (n_1)
- Information Gain is IG_1
- At second split,
- No. of data points be 500 (n_2)

- Information Gain is IG_{f_i}

\

We calculate feature importance of f_i by

<img

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This is called normalized information gain.

* We calculate normalized information gain for all features.

* This Normalized IG value is used as feature importances.

\

NOTE:

- This is just one of the popular way of calculating feature importance.

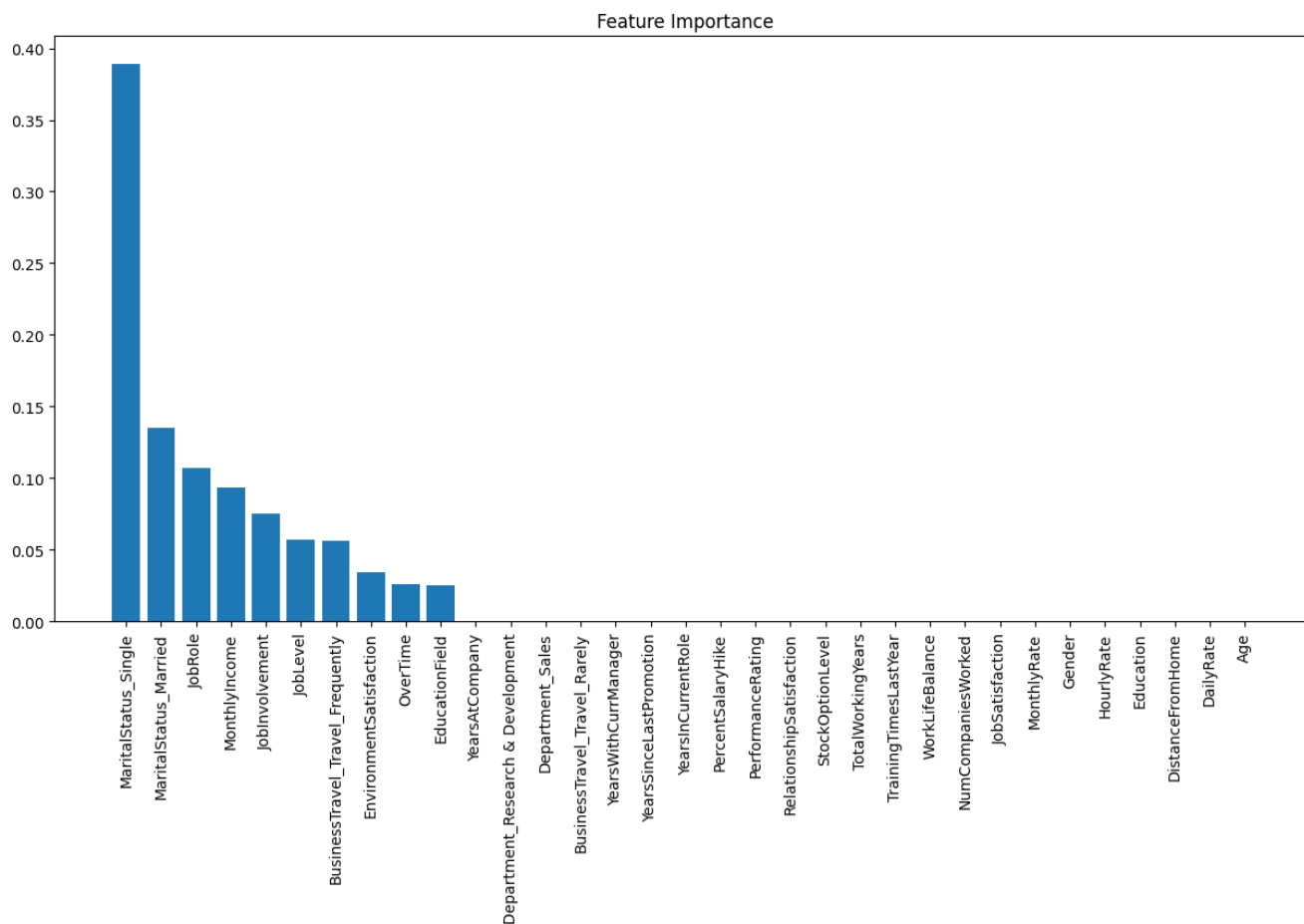
- Different libraries may use different methods of calculating it.

Code Walkthrough

Code

```
importances = tree_clf.feature_importances_  
indices = np.argsort(importances)[::-1] # Sort feature importances in descending order  
names = [X_train.columns[i] for i in indices] # Rearrange feature names so they match the  
sorted feature importances  
plt.figure(figsize=(15, 7)) # Create plot  
plt.title("Feature Importance") # Create plot title  
plt.bar(range(X_train.shape[1]), importances[indices]) # Add bars  
plt.xticks(range(X_train.shape[1]), names, rotation=90) # Add feature names as x-axis labels  
plt.show() # Show plot
```

Output Image



Interview questions

Important Terminologies

* The deeper you go building a tree that means you're reducing the no. of elements in your leaf node. This is known as Deep Tree. This leads to an overfit condition.

* Let's say if we have only one node. It means the depth of the tree is only 1 level. This is known as Decision Stump. Then this will result to an underfit condition.

* When we have a tree with smaller depth, we call it as a Shallow Tree. Note that a shallow tree has very less decision boundaries.

</br>

Q. Where do you think the impact of an outlier will be higher?

A deep tree or A shallow tree

The impact of an outlier will be lower on a shallow tree since here in the leaf node, we will have multiple values.

But in case of a deep tree, we might end up having only the outlier value present in the leaf node.

</br>

Q. What if we have imbalanced data? Will the DT be impacted by it?

Yes, the decision tree will surely be affected by the imbalanced dataset.

Say,

- $y_+ = 99\%$

- $y_- = 1\%$

We could use Oversampling/Undersampling methods or SMOTE technique that we learnt earlier.

</br>

Q. Does a DT requires standardised data?

Nope!

* Imagine we have a numerical feature x_i and we standardize it.

<img

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* But, does Impurity depends on values? \

Nope!

it depends on number of +ve/ -ve datapoints and in turn depends upon the threshold

\

- So standardisation won't change/affect the threshold.

- Hence, standardization doesn't impact the Entropy & IG.

Q. How much space is required to keep a DT in memory? How much time itll require to run?

Suppose,

- m is the no. of nodes in the decision tree.
- d is the depth of the decision tree.
- f is the no. of features present in the data.

Then,

- Space complexity - $O(m)$
- Runtime Complexity - $O(d)$
- Build time - $O(n \log n * f)$

</br>

Q. Can we use a DT for multi-class classification?

Yes, we surely can.

Suppose at the leaf node, we get the no. of values belonging to either one of the three given labels as -

* $y_1 = 300$

* $y_2 = 20$

* $y_3 = 10$

We can say that the sample belongs to the y_1 class.

</br>

Q. Can we use the same column for multiple splits?

Let us see 3 different cases for this -

</br>

Case-1 : NO </br>

</br>

Case-2 : YES </br>

</br>

\$Case-3\$: YES </br>

</br>

Q. If we were using a feature in multiple splits, how do we determine which feature is more important?

</br>

</br>

Q. How to use decision trees for Regression?

Lets understand this with an example

Suppose we have data of marks scored by students

- along with some bunch of categorical/ numerical features (\$f_1, f_2, f_3, f_4\$)

<img

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Using this data, we want to predict the marks.

Remember that,

- in regression problem, target variable (marks) is continuous.

So, how does decision tree handle continuous target variable ?

In order to so, lets breakdown the process DT follows:

- It calculates the impurity
- Splits the node to reduce the impurity
- Makes the predictions i.e. calculate predicted value

In short, we need to find answers to following questions:

- * How do we calculate impurity in case of regression ?
- * How do we calculate predicted value at each node ?

Lets try answering these questions

How to calculate impurity ?

How did we calculate impurity in case of classification?

We did so by calculating entropy/gini index.

Can we use the same over here as well?

* We cant. Why? coz both of these involve calculating probabilities
- and regression doesnt have probabilities

What shall we use instead ?

Instead, we use MSE

Now, in order to calculate MSE, we need two things

- Actual target values (we already have these)
- predicted target values (we dont have these)

which brings us to our next question

How to calculate predicted value ?

...

QUIZ 8:

What do you think DTs approach for calculating predicted value for regression?

- A. Majority vote
- B. Take mean of all values present at given node.
- C. Select random value out of present target values at given node.
- D. Takes first present values as predicted value

ANS: B. Take mean of all values present at the given node.

...

Decision Trees makes the prediction by simply

- taking average of all values present at node. i.e. mean value.

<img

src=https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/041/107/original/Screenshot_2023-07-31_at_10.30.24_AM.png?1690781264 width=700>